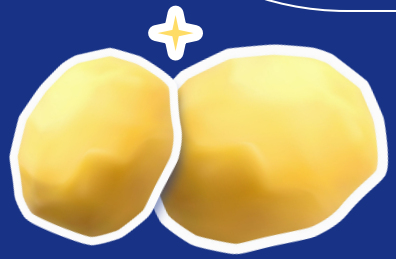
Tech Warmup II

TEAM 11 | POTATOES



CSE 110 SP26

Domain Research

PSYCHOLOGY

Color, Animation, Near-Misses, Unpredictable Rewards

LEGALITY

Specific RNG Algorithms, Age Verification

MECHANICS

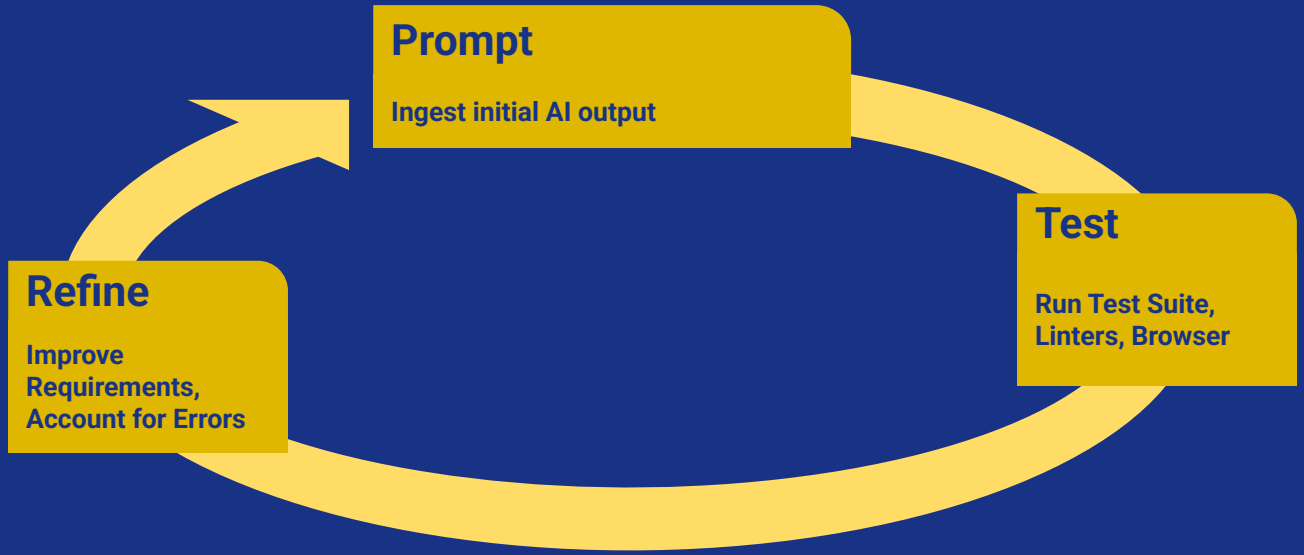
Reels & Paylines, Near-Misses, Payouts & Currency, Dual Mode

USERS

Multiple User Profiles, Maximise Appeal Across Demographics

Setup and Approach

- `gemini-3-flash-preview` in `gemini` CLI
- Maintained `ai-plan.md` file for persistence across iterations



What Worked

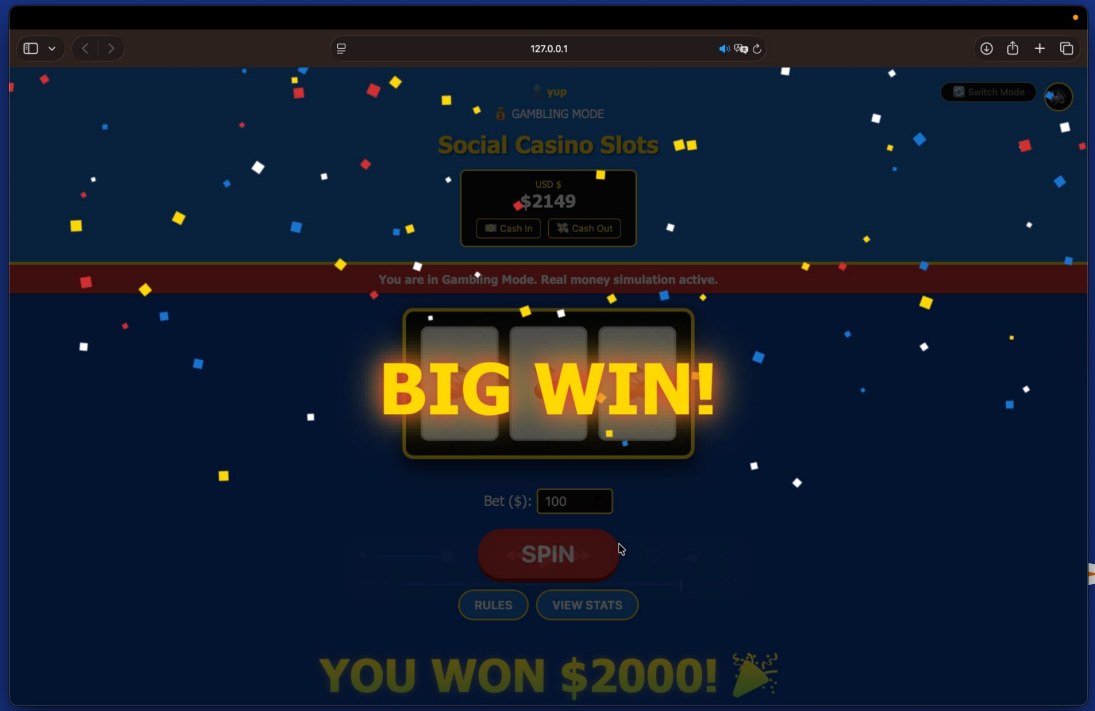
- Realistic RTP Engine
  - Cryptographically secure RNG with weighted reel math for accurate Return to Player simulation
- Virtual Economy
  - Functional wallet with a "Bankruptcy Rescue" feature so players never have to stop
- Progression Systems
  - XP/leveling and daily login streaks with bonus rewards
- Player Statistics
  - Tracks spins, biggest win, and total coins earned across sessions
- Flashy Visuals
  - Pure CSS/Vanilla JS reel animations and win confetti

What Didn’t Work

- Bugs and Fine Tuning required lots of iterations and passes
  - Especially in implementing (Payouts, Audio) or Refactoring
- Lots of junk
  - Each iteration generated voluminous code, necessity unknown
- Collaboration
  - Age-old problems like “works on my machine”
- Blind Spots
  - Tests passed but functionality still might be broken

Key Takeaways

- Prompt quality directly impacts output quality
- AI works best for small, incremental tasks
- AI-generated code often requires debugging and validation
- Clear planning improves AI effectiveness
- AI is a tool that supports developers, not a replacement

Final Reflection

Using AI is like gambling; you're not guaranteed an output, and your chances of winning are better if you know what you're doing.